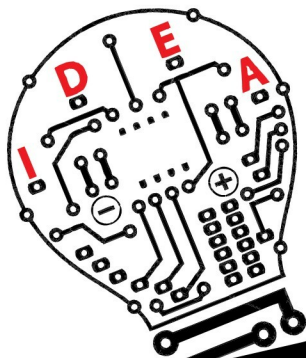


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offset, an example is shown in Fig. 4. Note the gain and offset values. Input parameters to be fed by you are $V_{ref} = 5$, $V_{outfs} = 4$, $V_{outzs} = 5$, $V_{infs} = -0.05$ and $V_{inzs} = -0.1$.

In this case, you have to enter suitable values for R_f and R_2 , and the rest are automatically calculated for the circuit design.

For negative gain and negative offset, an example is shown in Fig. 5. Input parameters are $V_{ref} = 5$, $V_{outfs} = 3$, $V_{outzs} = 0.2$, $V_{infs} = -0.2$ and $V_{inzs} = -0.1$.

Here, enter the value for R_f to get suitable circuit design values.

All relationships are based on the basic op-amp theory of biasing component value calculations. Resistor values are in ohms and voltage values are in volts.

Benefit of this program is that it displays a recommended design. As a beginner, you can benefit from this program as it may provide hands-on calculations for all op-amp based experiments.

Software installation and testing

1. Download WampServer (for localhost development) from www.wampserver.com/en/ and NetBeans IDE from <https://netbeans.org/features/php/>. Install these in your Windows PC. WampServer version 2.5 and NetBeans version 8.1 were used during testing at EFY Lab.

Install appropriate Visual C++ extension (here VC++ 2012) before installing WampServer to get all prerequisite .dll files for smooth running of the IDE and Apache server environment.

2. WampServer runs in the background with the option in online mode. Make sure WampServer icon on the taskbar turns green.

3. Create a folder, say, op-amp under C:\wamp\www folder. Copy opampCircuit.php, opampDesign.js, and opampRequirementForm.html files including image files into the op-amp project folder. (Note: These files have been included in this month's EFY DVD.)

4. Create a new PHP project in NetBeans. Select PHP Application option and click Next. A project folder will form automatically. Note that your op-amp folder is under this project folder.

5. Under Run Configuration window, make sure you select local server option in Run As: field; then, click OK to proceed.

6. Run opampRequirementForm.html file. You can feed various inputs as per your requirement and observe the outputs as shown in Figs 1 through 5. **EFY**

Ronit Adhiraj Ghosh is M.Tech from Homi Bhabha National Institute, Mumbai. His hobbies include electronics projects, watch collection, camping and hiking